

Why Measure Theory

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1 Introduction: Motivation

1.1 Definitions

I will begin with the definition of Riemann Integration.

Definition 1 (Partition). Let $[a, b]$ be the domain of a function then P a partition is of the form

$$P = \{x_0, x_1, \dots, x_n\}$$

Such that

$$a = x_0 < x_1 < \dots < x_n = b$$

Definition 2 (Upper sums). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$, the upper sums for f

$$U(f, P_\epsilon) = \sum_{k=1}^n (x_k - x_{k-1}) \sup_{[x_{k-1}, x_k]} f$$

Where $U(f)$ is defined as

$$U(f) = \inf_P U(f, P)$$

Definition 3 (Lower sums). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$, the lower sums for f

$$L(f, P_\epsilon) = \sum_{k=1}^n (x_k - x_{k-1}) \inf_{[x_{k-1}, x_k]} f$$

Where $L(f)$ is defined as

$$L(f) = \sup_P L(f, P)$$

Definition 4 (Reimann Integration). Let f be a bounded function $f: [a, b] \rightarrow \mathbb{R}$.

If for every $\epsilon > 0$ there exists a partition P_ϵ such that

$$U(f, P_\epsilon) - L(f, P_\epsilon) < \epsilon \quad (1)$$

then $f(x)$ is Riemann Integrable with

$$\int f = U(f) = L(f) \quad (2)$$

1.2 Riemann Failure

1.2.1 Function

Let the sequence of rational numbers $\{r_1, r_2, \dots\}$ enumerate the rationals on $[0, 1]$.

Let $f_n(x)$ be a series of functions of the form

$$f_n(x) = \begin{cases} 1 : & x \in \{r_1, r_2, \dots, r_n\} \\ 0 : & x \notin \{r_1, r_2, \dots, r_n\} \end{cases} \quad (3)$$

1.2.2 Upper and Lower Sums

Fix $n \in \mathbb{N}$.

Take the partition P_n such that $(x_k - x_{k-1}) = \frac{\epsilon}{4n}$ for all consecutive pairs of points in P_n .

The total number of intervals in which there are rational numbers is $2n$; since there are n rationals where each can be in at most 2 intervals at once.

Note. if $r_k = x_k$ for instance then $[x_{k-1}, x_k]$ and $[x_k, x_{k+1}]$ both contains r_k

This gives

$$U(f_n, P_n) = \left(\frac{\epsilon}{4n}\right)(2n)(1) + (1)(0) = \frac{\epsilon}{2}$$

Where each length $(x_k - x_{k-1}) = \left(\frac{\epsilon}{4n}\right)$, there exists at most $2n$ of these intervals, each with $\sup_{[x_k - x_{k-1}]} f = 1$. All other possible intervals have a max combined length of 1, each with $\sup_{[x_k - x_{k-1}]} f = 0$.

Since $\mathbb{I}$ is dense in $\mathbb{R}$, between any two real numbers there exists an irrational. Meaning on any interval $[x_{k-1}, x_k]$, $\inf_{[x_{k-1}, x_k]} f = 0$ giving

$$L(f_n, P_n) = 0$$

This gives for any $\epsilon > 0$ for some $N \in \mathbb{N}$ whenever $n \geq N$ it follows

$$U(f_n, P_n) - L(f_n, P_n) = \frac{\epsilon}{2} < \epsilon$$

meaning the function is Riemann integrable with

$$\int_0^1 f_n = L(f_n) = 0$$

1.2.3 Limit

Since the sequence of values r_k are countable we have

$$\lim_{n \rightarrow \infty} f_n = \begin{cases} 1 : & x \in \mathbb{Q} \cap [0, 1] \\ 0 : & x \notin \mathbb{Q} \cap [0, 1] \end{cases} \quad (4)$$

Call this function f

Now for any partition P , since both $\mathbb{Q}$ is dense in $\mathbb{R}$ as well as $\mathbb{I}$ dense in $\mathbb{R}$, on any interval $[x_{k-1}, x_k] \in P$ there exists both a rational and irrational.

This gives $\inf_{[x_{k-1}, x_k]} f = 0$ as well $\sup_{[x_k - x_{k-1}]} f = 1$ on every interval. Which forces

$$U(f, P) = 1$$

and

$$L(f, P) = 0$$

for every partition P implying $U(f) = 1$ and $L(f) = 0$ which says that f is not Riemann integrable.

1.3 Lebesgue Explanation

1.3.1 Measure 0

Definition 5 (Measure 0). A is a measure zero set if for every $\epsilon > 0$ there exists a countable covering $\bigcup_n O_n$ where each O_n is an open interval such that

$$A \subset \bigcup_n O_n \quad (5)$$

Where

$$\sum_n l(O_n) \leq \epsilon \quad (6)$$

and $l()$ denotes the length of a open interval defined as $b - a$ for any open interval (a, b) .

1.3.2 Lebesgue Criterion

By the Lebesgue criterion

Theorem 1 (Lebesgue Criterion). *A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable iff the set of discounties is Measure 0*

1.3.3 f discontinuous everywhere

Recall that

$$\lim_{n \rightarrow \infty} f_n = \begin{cases} 1 : & x \in \mathbb{Q} \cap [0, 1] \\ 0 : & x \notin \mathbb{Q} \cap [0, 1] \end{cases}$$

I want to show that f is discontinuous on all of $[0, 1]$ First observe the definition of continuity.

Definition 6. A function f is continuous at a point c if for every $\epsilon > 0$ there exists a $\delta > 0$ such that whenever $|x - c| < \delta$ it follows that $|f(x) - f(c)| < \epsilon$

Now fix $\epsilon_0 \leq 1$ take $\epsilon_0 = \frac{1}{2}$. Choose any point $c \in [0, 1]$. Again both $\mathbb{Q}$ and $\mathbb{I}$ dense in $\mathbb{R}$.

First consider $c \in \mathbb{Q}$ then for any δ chosen there exists some irrational point

$$x \in (c - \delta, c + \delta) \implies |f(x) - f(c)| = 1 \geq \epsilon_0 \quad (7)$$

Consider the case $c \in \mathbb{I}$ then for any δ chosen there exists some rational point

$$x \in (c - \delta, c + \delta) \implies |f(x) - f(c)| = 1 \geq \epsilon_0 \quad (8)$$

Meaning the set of discontinuous points of f is all of $[0, 1]$ where $|[0, 1]| = 1$.

Note. I do not prove it here but it is proved in section 2.2

To show that the set of discontinuities for every f_n is measure 0 first consider the set of points at which f_n is discontinuous, call it D_n .

For any points $c \in \{r_1, r_2, \dots, r_n\}$ since $\mathbb{I}$ dense in $\mathbb{R}$ we have for a fixed $\epsilon < 1$ and any $\delta > 0$

$$|x - c| < \delta \implies |f(x) - f(c)| = 1 \geq \epsilon$$

Making

$$D_n = \{r_1, r_2, \dots, r_n\}$$

For any points $c \notin \{r_1, r_2, \dots, r_n\}$ we can take $\delta = \frac{\min(|x-r_1|, |x-r_2|, \dots, |x-r_n|)}{2}$ since there are a fixed amount of rationals for any fixed $n \in \mathbb{N}$.

This would give us for

$$|x - c| < \delta \implies |f(x) - f(c)| = 0 < \epsilon$$

making f_n continuous on any points $x \notin \{r_1, r_2, \dots, r_n\}$.

Now consider the open covering $\bigcup_k O_k$ such that each

$$O_k = (r_k - \frac{\epsilon}{2^{k+1}}, r_k + \frac{\epsilon}{2^{k+1}})$$

Then

$$\begin{aligned} \sum_k l(O_k) &= \sum_k \frac{\epsilon}{2^k} \\ &= \frac{\frac{\epsilon}{2}}{1 - \frac{1}{2}} \\ &= \epsilon \end{aligned}$$

giving

$$\sum_k l(O_k) \leq \epsilon$$

making D_n measure 0 by definition.

We now understand why each f_n integrates but f itself fails to integrate. Our understanding lays not in the language of the Riemann integral itself but instead in the language of measure.

We also see that the method by which he partition the domain fails for functions that oscillate too quickly and will fail to be defined in spaces outside of $\mathbb{R}$

1.4 Lebesgue Savior

2 Outer Measure

Definition 7 (Outer Measure). $|A| = \inf \sum_k l(O_k)$ Where $\inf$ is taken over all open coverings such that $A \subset \bigcup_k O_k$

2.1 Agrees everywhere

Outer measure agrees with length on open sets and $[a, b] = (a, b)$ in terms of measure. Translation invariance, subadditivity, monotonicity

- Theorem 2.**
1. If $A \subseteq B$ then $|A| \leq |B|$
 2. $|[a, b]| = b - a$
 3. If $A_1, A_2, \dots$ satisfy $\bigcup_k A_k = B$, then $|B| \leq \sum_k |A_k|$
 4. $|A + t| = |A|$

I will now prove each in order.

Proof: Monotonicity.

For monotonicity (1) first observe that since

$$A \subset B \subset \bigcup_k O_k$$

any open covering of B is an open covering of A .

Now outer measure is defined as inf over all open coverings. Therefore there exists an open covering $\bigcup_k O_k$, such that

$$\bigcup_k l(O_k) < |B| + \epsilon$$

Since any open covering of B is an open covering of A

$$|A| \leq \sum_k l(O_k) < |B| + \epsilon$$

Where ϵ is arbitrary > 0 . Meaning $|A| \leq |B|$

□

Proof: $|[a, b]| = b - a$.

For the second property of outer measure consider the open cover for $[a, b]$

$$(a - \epsilon, a + \epsilon) \cup (a, b) \cup (b - \epsilon, b + \epsilon)$$

Since $|[a, b]|$ is inf over all coverings we have

$$|[a, b]| \leq 2\epsilon + b - a + 2\epsilon$$

Where ϵ is arbitrary giving

$$|[a, b]| \leq b - a$$

For the other direction consider any open covering of $[a, b]$. I want to show that

$$\sum_k l(O_k) \geq b - a$$

hence giving

$$\inf \sum_k l(O_k) \geq b - a$$

Now $[a, b]$ is a compact interval, by Heine-Borel for every open cover

$$[a, b] \subset \bigcup_k O_k$$

there exists a finite subcollection such that

$$[a, b] \subset \bigcup_{k=1}^n O_k$$

Where

$$\sum_{k=1}^n l(O_k) \leq \sum_{k=1}^{\infty} l(O_k)$$

Since length is non negative.

Suppose the claim holds for n intervals.

Begin with $n = 1$ intervals. Any interval containing $[a, b]$ ie $[a, b] \subset O_k$ has length $l(O_k) \geq b - a$, since

$$(a, b) \subset [a, b] \subset O_k$$

Now consider $n + 1$ intervals covering $[a, b]$.

b must be contained in at least one of these intervals call it (c, d) . If $c < a$ then $l(c, d) \geq b - a$ so consider $a < c < b < d$; since (c, d) covers none of $[a, c]$ we can invoke the inductive hypothesis

$$\sum_{k=1}^n l(O_k) \geq c - a$$

for O_k covering $[a, c]$. Adding this to $l(c, d)$ gives

$$\sum_{k=1}^{n+1} l(O_k) \geq (c - a) + (d - c)$$

$$= d - a$$

$$\geq b - a$$

Completing the proof. □

Proof: Countable Subadditivity.

I want to show

$$|\bigcup_k A_k| \leq \sum_k |A_k|$$

Let

$$A = \bigcup_k A_k$$

Since outer measure is inf over all open covers, take for each A_k the open cover

$$O_k < |A_k| + \frac{\epsilon}{2^{k+1}}$$

Now since the O_k cover A

$$A \subset \bigcup_k O_k$$

giving

$$\begin{aligned} |A| &\leq |\bigcup_k O_k| \\ &= \sum_k |O_k| \\ &= \sum_k \frac{\epsilon}{2^{k+1}} \\ &= \frac{\epsilon}{2} \end{aligned}$$

Since this is true for any arbitrary ϵ and A is inf over all coverings we get

$$|\bigcup_k A_k| \leq \sum_k |A_k|$$

Completing the proof.

□

Proof: Translation Invariance.

I would like to show $|A + t| = |A|$.

First observe that

$$l((a, b)) = l((a + t, b + t))$$

since

$$l((a + t, b + t)) = (b + t) - (a + t) = b - a$$

Now observe that for any open cover of A call it O_A , O_{A+t} covers A .

Where O_{A+t} is defined as

$$\bigcup_k (O_k + t)$$

for $O_k \in O_A$.

Take any point in $A + t$ call it y . Then $y = x + t$ for some $x \in A$. Where $x \in O_k$ for some $O_k \in O_A$. Then $y \in O_k + t$. Where

$$l(O_k) = l(O_k + t)$$

Making the set of covers of A , call them $S \subset S_t$, where S_t is the set of open covers of $A + t$.

Meaning since $|A|$ inf over all covers ie $\inf S$ and $S \subset S_t$, it follows $|A + t| \leq |A|$.

Note. Say $B \subset C$ then $\inf C \leq \inf B$; since taking inf over larger set is smaller.

Now consider the sets $A+t+(-t)$ and $A+t$ where the left side is a translations of the right side. Giving

$$|A + t + (-t)| \leq |A + t| \implies |A| \leq |A + t|$$

Which follows from the above. Completing the proof $|A + t| = |A|$.

□

It seems that outer measure has all the desired properties we would want or need. But we would want to see that in general for disjoint sets

$$|\bigcup_k A_k| = \sum_k |A_k| \tag{9}$$

But as we will see in a moment this is not always the case.

2.2 Failure of Outer Measure as a Measure

2.2.1 Equivalence Classes

Definition 8 (Equivalence Relation). x is said to be equivalent to y denoted by $x \sim y$ iff $x - y \in \mathbb{Q}$

To be an equivalence relation it must satisfy $x \sim x$, if $x \sim y$ then $y \sim x$, and if $x \sim y$ and $y \sim z$ then $x \sim z$.

Proof: Equivalence Relation.

For the first, notice for any value x , $x - x = 0$ and $0 \in \mathbb{Q}$.

For the second, notice that $y - x = -1(x - y)$ where $\mathbb{Q}$ is closed under multiplication.

For the third, notice $(x - y) + (y - z) = (x - z)$ where $\mathbb{Q}$ is closed under addition. $\square$

Definition 9 (Equivalence Class on $[0, 1]$).] An equivalence class on $[0, 1]$ is a set of all equivalent elements $y : y \sim x$ for all $x \in [0, 1]$

Now any two equivalence classes on $[0, 1]$ are disjoint. To see this suppose the intersection of two equivalence classes was non empty. So let there be some element in A call it z and the same element in B . By definition $x - z \in \mathbb{Q}$ and $y - z \in \mathbb{Q}$ for arbitrary $x \in A$ and $y \in B$. Let $x - z = u$ and $y - z = v$. Solving for x and y gives $x = u + z$ and $y = v + z$. Now if we subtract $x - y$, we get $x - y = (u + z) - (v + z) = u - v$. But $u, v \in \mathbb{Q}$ where $\mathbb{Q}$ is closed under subtraction giving $x - y \in \mathbb{Q}$, but that says $x \sim y$.

Then taking any $x \in A$ gives $x \sim y$ for some $y \in B$ forcing $x \in B$. Making $A \subset B$. Similarly taking any $y \in B$ gives $y \sim x$ for some $x \in A$. Making $B \subset A$ or $A = B$. Therefore any two equivalence classes on $[0, 1]$ are disjoint.

2.2.2 Vitali

Let V contain a single element from each equivalence class. There exists uncountably many of these classes in which we have no way of choosing, but where we can still choose an element from each from the Axiom of Choice.

Now for every $q \in [-1, 1] \cap \mathbb{Q}$ the sets $V + q$ are disjoint. To see this suppose some element w existed such that $w \in (V + p) \cap (V + q)$ for $p, q \in \mathbb{Q} \cap [0, 1]$. Then $w = v + p$ for some $v \in V$ and $w = u + q$ for some $u \in V$. So $v - u = q - p$ making $v - u \in \mathbb{Q}$ meaning $v \sim u$ but V only contains one element from each equivalence class meaning $v = u$ forcing $p = q$. But that says the translated classes are the same, meaning each translated class must be disjoint.

Now

$$[0, 1] \subset \bigcup_q (V + q) \subset [-1, 2] \quad (10)$$

Take any arbitrary $x \in [0, 1]$, x sits in some equivalence class where V contains some representative from that class. Giving $x - v \in \mathbb{Q}$. Since $x, v \in [0, 1] \implies x - v \in [-1, 1] \cap \mathbb{Q}$. Where every $q \in [-1, 1]$ is used. Now take an arbitrary $v + q$ simply $v \in [0, 1]$ and $q \in [-1, 1]$ forcing $v + q \in [-1, 2]$.

Now by monotonicity it must be the case that

$$|[0, 1]| \leq \left| \bigcup_q (V + q) \right| \leq [-1, 2] \quad (11)$$

Which implies

$$1 \leq \left| \bigcup_q (V + q) \right| \leq 3 \quad (12)$$

Since each $V + q$ consists of exactly the same amount of elements $|V + q| = |V + p|$.

Now suppose it is true for disjoint sets

$$\left| \bigcup_k A_k \right| = \sum_k |A_k| \quad (13)$$

then

$$\left| \bigcup_q (V + q) \right| = \sum_q |(V + q)| \quad (14)$$

Suppose that $|(V + q)| > 0$, since there are an infinite amount of $q \in [-1, 1]$ it would force $\sum_q |(V + q)| = \infty$.

Suppose then that $|(V + q)| = 0$, it would then force $\sum_q |(V + q)| = 0$.

But measure is non negative meaning $|(V + q)| \geq 0$ but ether gives a contra-

diction to

$$1 \leq |\bigcup_q (V + q)| \leq 3 \quad (15)$$

Meaning for the vitalli set, (a valid subset of $\mathbb{R}$ implying the axiom of choice)

$$|\bigcup_k A_k| = \sum_k |A_k| \quad (16)$$

does not hold.

Meaning it is not true in general and there we can not assert that

$$|\bigcup_k A_k| = \sum_k |A_k| \quad (17)$$

This as you will see later actually implies that outer measure is not a valid measure on all subsets of $\mathbb{R}$.

3 Integration on Measure

We can now generalize to any space (x, S, mew) and we have measure which splits based on output and reverse maps to inputs where we get the the measure of those sets times the inf of the function on the sets, which we can then use for probability where we know for certain our methods are coherent.